

Andrii Kulbaba

ROBOTICS & BACKEND ENGINEER

Lviv, Ukraine

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“Be the change that you want to see in the world.”

Summary

Sophomore student of the Robotics CS program at the Ukrainian Catholic University. My vision is centered on the future of engineering: designing autonomous robotic systems capable of exploring land, air, and deep space. I strive to bridge the gap between rigorous computational thinking and the massive data streams generated by sensors on autonomous explorers. With a technical background in embedded C++, computer vision, and backend orchestration, I am focused on building resilient machines that can autonomously navigate and map the challenges of extraterrestrial environments.

Skills

Programming	C++, Python, SQL, TypeScript, Go
Backend & Databases	FastAPI, PostgreSQL, SQLAlchemy, Pydantic, JWT, OAuth2
Frontend & Design	React, Figma, UI/UX Prototyping
Embedded & Robotics	ESP-IDF, FreeRTOS, ESP32-S3, ESP32-Camera, Computer Vision
DevOps & Tools	Docker, Docker Compose, Git, Linux Terminal
Mathematics & Algorithms	Calculus, Discrete Math, Data Compression (LZ77, LZW, DEFLATE) Algorithm Theory

Technical Experience

UCU @ Web Development Course

BACKEND & DEVOPS ENGINEER

Lviv, Ukraine

Dec. 2025 - Jan. 2026

- Engineered a high-performance RESTful API for “Linknet” bookmark manager using the FastAPI framework.
- Optimized a relational database schema in PostgreSQL using SQLAlchemy ORM for User-Category-Tab hierarchies.
- Implemented secure authentication using JWT and OAuth2 with PBKDF2 password hashing.
- Containerized the entire stack (React, FastAPI, PostgreSQL) using Docker Compose for environment consistency.
- Developed robust data validation layers using Pydantic and implemented cascading deletes for data integrity.

Ground Swarm Robotics Project

EMBEDDED SYSTEMS & ROBOTICS RESEARCHER

Lviv, Ukraine

Sep. 2025 - Present

- Designed and prototyped a fleet of ground-based robots for testing swarm intelligence and flocking algorithms.
- Developed embedded firmware in C++ using ESP-IDF, leveraging FreeRTOS for dual-core task scheduling on ESP32-S3.
- Implemented a computer vision system for robot recognition and visual tracking via ESP32-Camera.
- Engineered a low-latency wireless control system via Bluetooth for manual joystick intervention and real-time telemetry.
- Integrated sensor fusion logic to ensure stable physical movement during swarm algorithm validation.

UCU @ Fluxon Bootcamp

PROJECT LEAD & FRONTEND DEVELOPER

Lviv, Ukraine

Jul. 2025

- Conceptualized and led the development of a full-stack crypto monitoring application for cross-ticker calculations.
- Designed high-fidelity UI/UX mockups in Figma with a focus on intuitive financial data visualization.
- Developed the core frontend architecture using React, ensuring seamless interaction with real-time crypto APIs.
- Integrated Firebase for cloud storage and secure authentication, managing persistent user settings.
- Coordinated the project lifecycle as a PM, bridging the gap between design requirements and technical implementation.

UCU @ Discrete Mathematics II Project

PROJECT LEAD & BACKEND DEVELOPER

Lviv, Ukraine

Apr. 2025 - May. 2025

- Led a team of 4 in researching data compression and developing a cross-platform codec suite in Python.
- Engineered high-performance implementations of LZ77, LZW, and DEFLATE algorithms.
- Developed a specialized DPCM codec for lossless .wav audio signal processing.
- Conducted an empirical study on compression ratios and execution speeds across various data types.

Delph Project

SOFTWARE ARCHITECT & BACKEND DEVELOPER

Lviv, Ukraine

Mar. 2025 - May. 2025

- Architected a full-stack monitoring system for high-performance data management using FastAPI and React.
- Integrated Apache Kafka to enable fast, asynchronous event-driven communication between system components.
- Designed a robust PostgreSQL database, optimizing schema structures for high-frequency data retrieval.
- Orchestrated the infrastructure using Docker, simplifying deployment pipelines and development workflows.

UCU @ Discrete Mathematics I Project

PROJECT LEAD & CORE ALGORITHM DEVELOPER

Lviv, Ukraine

Nov. 2024 - Dec. 2024

- Led the development of an Ant Colony Optimization (ACO) algorithm to solve the TSP and pathfinding on complex graphs.
- Implemented probabilistic state transitions and pheromone management models for evaporation and reinforcement.
- Optimized performance using Multiprocessing, achieving significant speedups via multi-core iteration distribution.
- Validated the solver on 15-node complete graphs, achieving optimal results in under 20 seconds.

Education

UCU(UCU APPS)

B.S. 2ND YEAR IN COMPUTER SCIENCE AND ENGINEERING

Lviv, Ukraine

sep. 2024 - Jun. 2028

- Got a Fluxon Certification during the summer school of web development

Extracurricular Activity

Student Sport Organization at UCU

FINANCE OFFICER

Lviv, Ukraine

Sep. 2025 - Present

- Acted as a primary financial liaison between various departments, processing internal reports and assessing resource requirements.
- Developed and implemented financial strategies to secure funding for organizational initiatives and sports events.
- Coordinated communication with government bodies and external sports organizations to facilitate partnership opportunities.
- Proposed and executed marketing activities to increase organizational awareness and expand the professional network.

UCU Collegium

WING HEAD & PROJECT LEAD

Lviv, Ukraine

Sep. 2025 - Present

- Led a student wing in the UCU Collegium, fostering a strong community network and promoting the university's ethos of service.
- Managed a team of 30+ people during the strategic planning and execution of large-scale community events.
- Directed a humanitarian and military volunteer project, coordinating a team of 10 to provide crisis relief and logistics support.
- Mentored students through social responsibility projects focused on assistance for individuals with disabilities.